THAKUR COLLEGE OF ENGINEERING & TECHNOLOGY
(Approved by AICTE, Govt. of Maharashtra & Affiliated to University of Mumbai)
A - Block, Thakur Educational Campus, Shyamnarayan Thakur Marg,
Thakur Village, Kandivali (East), Mumbai - 400 101.

IN-SEMESTER EXAMINATION-II

F.E./F.T. (Semester-I)

SUBJECT – Programming for Problem Solving

Branch: AIML, E&TC, E&CS, IT, MECH, MME **Date: 23/12/2025**
Div.: ALL **Timing: 12:00 PM to 1:00 PM**
Duration: 60 Minutes **Maximum Marks: 20**

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	What is the significance of function parameters in C?	2	CO4	U
	b.	Show advantages & disadvantages of recursion.	2	CO4	A
	c.	State & explain Array. State different ways of initialization of the same.	2	CO5	R
	d.	Illustrate String. State different ways of initialization of the same.	2	CO5	A
	e.	What is meant by array of structures?	2	CO6	U
	f.	Write the syntax to declare a pointer variable.	2	CO6	R
	g.	What is the use of the address-of operator (&)?	2	CO6	R
Q.2	a.	Explain Storage Classes in detail with example. (a)Static (b)Extern	5	CO4	R
		OR			
	b.	Write a program to add two matrices (2D arrays) in C.	5	CO5	U
Q.3	a.	State all the string library functions and Write a program to calculate length of a string without using in-built functions.	5	CO5	R
		OR			
	b.	Explain dynamic memory allocation functions: malloc(), calloc() with example.	5	CO6	A